Homework 3

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1. Let G be a group and let H be a subgroup of G. Prove that $C_G(H) \subset N_G(H)$. Prove that equality holds if H has order two.

Proof. If $g \in C_G(H)$, then $ghg^{-1} = h$ for each $h \in H$, so $gHg^{-1} = H$. Thus $g \in N_G(H)$.

If H has order two, then one of the two elements of H is e, and the other is an element of order two, say x. If $g \in N_G(H)$, then $geg^{-1} = e$, and g must act on x to take it to the only remaining element of H, which is x itself, so $gxg^{-1} = x$. $\square$

2. Let G be a group, and suppose H is a finite subgroup of G generated by a set X. Prove that $g \in N_G(H)$ if and only if $gxg^{-1} \in H$ for all $x \in X$. Conclude that for $x \in G$, one has that $g \in N_G(\langle x \rangle)$ if and only if $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$.

Proof. If $g \in N_G(H)$, then gHg^{-1}, so indeed $gxg^{-1} \in H$ for any $x \in X \subset H$.

Now let $g \in N_G(H)$ and assume that $gxg^{-1} \in H$ for all $x \in X$. Then given $h \in H$, we can write $h = gh'g^{-1}$ for some $h' \in H$, so $h' = x_1^{a_1} \cdots x_n^{a_n}$ for some $x_i \in X$ and each $a_j \in \mathbb{Z}$. We then write

$$h = gh'g^{-1} = gx_1^{a_1}x_2a_2 \cdots x_n^{a_n}g^{-1} = (gx_1g^{-1})^{a_1}(gx_2g^{-1})^{a_2} \cdots (gx_ng^{-1})^{a_n},$$

so h is a product of elements of H. Thus $gHg^{-1} \subset H$.

Now assume that $h, h' \in gHg^{-1}$. Then

$$ghg^{-1} = gh'g^{-1} \implies h = h'.$$

Thus since H is finite and $f : H \to H$ defined by $f(h) = ghg^{-1}$ is injective, f is in fact a bijection, and $gHg^{-1} = H$. $\square$

3. Let $H = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle \subset \mathrm{GL}_2(\mathbb{R})$, and let $g = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. Show that $gHg^{-1} \subset H$, but that this containment is proper and hence g is *not* in the normalizer of H. Why does this not violate the statement in Problem 2?

Proof. Since any element of H is of the form $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ and $g^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$, we can calculate explicitly

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}.$$

Since this only includes matrices with an even upper-right entry, it excludes $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ itself, and thus $gHg^{-1} \subsetneq H$. This is in complete agreement with Problem 2, since Problem 2 required the subgroup in question, here H, to be finite, and H is infinite. $\square$

4. Let G be a group, and n a positive integer. Prove that the set of group homomorphisms from $\mathbb{Z}_n$ to G are in bijection with the set of elements of G of order dividing n.

Proof. We will write $\mathbb{Z}_n = \langle x : x^n \rangle$ for ease of notation. The idea here is that a homomorphism from $\mathbb{Z}_n$ to G is uniquely determined by where it sends x. Indeed, given a homomorphism φ which sends x to $g \in G$, then $\varphi(x^i) = \varphi(x)^i = g^i$, so φ is described completely by where it sends x since x generates $\mathbb{Z}_n$. A direct consequence is the injectivity we need. Then, using the fact that $|\varphi(x)|$ divides $|x| = n$, we determine that we can only send x to elements of G with order dividing that of x. In fact, given any $g \in G$ with $|g|$ dividing n, there is a homomorphism determined entirely by sending x to g, so we have surjectivity. $\square$

5. A permutation $\sigma \in S_n$ of the form $(i\ j)$ for $1 \leq i < j \leq n$ (in cycle notation) is called a *transformation.* If $j = i+1$, we say the transposition is *adjacent.* Prove that S_n is generated by the set of transpositions. Then prove that all transpositions are products of adjacent transpositions to conclude that S_n is in fact generated by adjacent transpositions.

Proof. First, we show that every element of S_n is a product of transpositions. Indeed, for a cycle $(a_1\ a_2\ \cdots\ a_m)$, we can write

$$(a_1\ a_2\ \cdots\ a_m) = (a_{m-1}\ a_m) \cdots (a_2\ a_m)(a_1\ a_m).$$

Given any transposition, then, we want to show that it is a product of adjacent transpositions, which we will prove by induction. We write any transposition as $(a\ [a+i])$. Let $\sigma \in S_n$. If $i = 0$, then σ is the identity permutation, which is the product of zero adjacent transpositions. If $i = 1$, Then $\sigma = (a\ [a+1])$ is an adjacent permutation by definition. Now assume that $i = n$ and the hypothesis holds for $i < n$. Then we can write

$$(a\ [a+n]) = (a\ [a+n-1])([a+n-1]\ [a+n])(a\ [a+n-1]),$$

and since $(a\ [a+n-1])$ can be written as a product of adjacent permutations, this represents $(a\ [a+n])$ as a product of adjacent permutations. $\square$

6. Let $p \geq 3$ be a prime integer. Prove that D_{2p} is generated by any pair of distinct reflections. Show by means of example that this can fail if p is not prime.

Proof. Choose a reflection, which we call s, and a rotation, which we call r. We can now choose a reflection distinct from s that can be written as $r^n s$ for some n.

First, we observe that since p is prime, the subgroup generated by r has order p, regardless which rotation we have chosen. We now want to show that s and $r^n s$ generate the group. Note that s has order 2, and since $(r^n s)(s) = r^n$ is an element of the subgroup generated by r, we have that $r^{nm} = r$ for some m. Thus we have produced both standard generators of D_{2n} (r and s), and all relations are satisfied ($r^p = s^2 = e$ together with the inherited $r^i s = sr^{-i}$ for any i). $\square$

7. Let G be a group and let H be a subgroup. Let G act on the set G/H by left translation. Prove that the kernel of this action is $K = \bigcap_{g\in G} gHg^{-1}$, and prove that K is the largest normal subgroup of G contained in H.

Proof. Let $k \in K$. First we show that $K \subset \bigcap_{g\in G} gHg^{-1}$. We want to prove that given $g \in G$, $k = ghg^{-1}$ for some $h \in H$, so we will show that $g^{-1}kg \in H$, since that would suffice for h. Since $k \in K$, $kgH = gH$, so $g^{-1}kgH = g^{-1}(kgH) = (g^{-1}g)H = H$. Thus, $g^{-1}kg \in H$ as desired. For the reverse containment, fix $g \in G$ and let $k = ghg^{-1}$ for some $h \in H$. Then $ghg^{-1}(gH) = gh(g^{-1}g)H = ghH = gH$, so since g was arbitrary, $ghg^{-1} \in K$.

Now let J be a normal subgroup of G contained in H, and let $j \in J$. Then given any $g \in G$, since $gJg^{-1} = J$, there exists a $j' \in J$ such that $gj'g^{-1} = j$. Since $j' \in J \subset H$ and the choice of g was arbitrary, we have

$$j \in \bigcap_{g\in G} gHg^{-1} = K \implies J \subset K.$$

$\square$

8. Suppose G acts on a set X, let $x \in X$, and let $H = \mathrm{Stab}_G(x)$. Recall that G/H denotes the set of left cosets of H in G. Prove that the assignment

$$\begin{aligned} \varphi : G/H &\to \mathrm{Orb}_G(x) \\ gH &\mapsto g \cdot x \end{aligned}$$

is a well-defined bijective function. Furthermore, show that φ is G-*equivalent*; that is, prove that for all $a \in G$ and all $gH \in G/H$, one has that $\varphi(a \cdot gH) = a \cdot \varphi(gH)$, where the G-action on G/H is left translation of cosets, and the G-action on $\mathrm{Orb}_G(x)$ is inherited from the action of G on X.

Proof. Let $g_1, g_2 \in G$ such that $g_1H = g_2H$. Then given $h_1 \in H$, there exists an $h_2 \in H$ such that $g_1h_1 = g_2h_2$. We can then deduce that

$$\begin{aligned} g_1 \cdot x &= g_1 \cdot (h_1 \cdot x) \\ &= (g_1h_1) \cdot x \\ &= (g_2h_2) \cdot x \\ &= g_2 \cdot (h_2 \cdot x) \\ &= g_2 \cdot x, \end{aligned}$$

so φ is a well-defined function.

By a similar line of argument, we can show injectivity. Assume $g_1H \neq g_2H$ for some $g_1, g_2 \in G$. Then g_1H and g_2H are disjoint, so given any $h_1, h_2 \in H$, $g_1h_1 \neq g_2h_2$. Then we have

$$\begin{aligned} g_1 \cdot x &= g_1 \cdot (h_1 \cdot x) \\ &= (g_1h_1) \cdot x \\ &\neq (g_2h_2) \cdot x \\ &= g_2 \cdot (h_2 \cdot x) \\ &= g_2 \cdot x. \end{aligned}$$

Finally, given any $g \cdot x \in \mathrm{Orb}_G(x)$, the coset gH maps to $g \cdot x$, so φ is a bijective function.

We conclude by showing G-equivariance:

$$\begin{aligned}\varphi(a \cdot gH) &= \varphi((ag)H) \\ &= (ag) \cdot x \\ &= a \cdot (g \cdot x) \\ &= a \cdot \varphi(gH).\end{aligned}$$

$\square$

9. Let $\varphi : G \to H$ be a group homomorphism. Prove that if N is a subgroup of H then $\varphi^{-1}(N)$ is a subgroup of G. Prove that if N is normal, so is $\varphi^{-1}(N)$.

Proof. Let $g, h \in \varphi^{-1}(N)$. Then $\varphi(gh^{-1}) = \varphi(g)\varphi(h)^{-1} \in N$, so $gh^{-1} \in \varphi^{-1}(N)$.

Now given $g \in G$ and $n \in \varphi^{-1}(N)$, we have $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} \in N$ since $\varphi(n) \in N$ and N is normal. Thus $gng^{-1} \in \varphi^{-1}(N)$, so since g and n were arbitrary elements of G and $\varphi^{-1}(N)$ respectively, $\varphi^{-1}(N)$ is normal. $\square$

10. Suppose that G is a group and that X is a subset of G that is closed under conjugation. Prove that $\langle X \rangle$ is a normal subgroup. Next, let Y be the set of commutators

$$\left\{aba^{-1}b^{-1} : a, b \in G\right\}$$

and prove that Y is closed under conjugation. Use these facts to show that the commutator subgroup $[G, G] = \langle Y \rangle$ is a normal subgroup.

Proof. If $n \in \langle X \rangle$ and $g \in G$, write $n = n_1 n_2 \cdots n_m$ where each $n_i \in X$. Then

$$gng^{-1} = gn_1 \cdots n_m g^{-1} = (gn_1 g^{-1}) \cdots (gn_m g^{-1}) \in \langle X \rangle$$

since $gn_i g^{-1} \in X$ for each i.

Now, given any $a, b, g \in G$, we have

$$g(aba^{-1}b^{-1})g^{-1} = (gag^{-1})(gbg^{-1})(gag^{-1})^{-1}(gbg^{-1})^{-1} \in Y.$$

Thus $[G, G] = \langle Y \rangle$ is a normal subgroup. $\square$